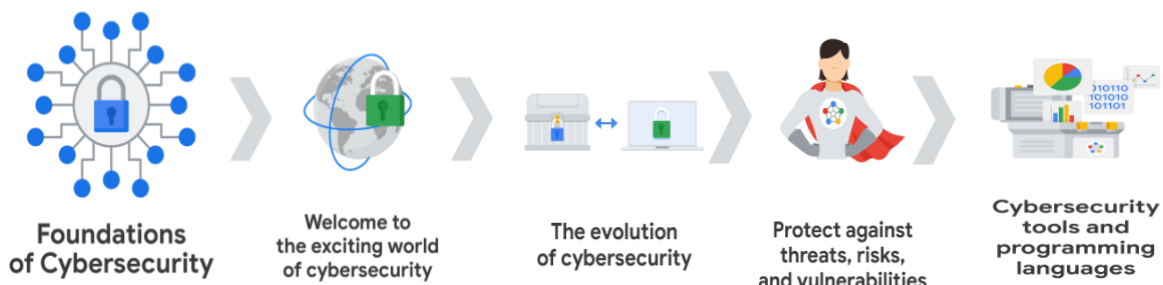


Course Overview

- **Module 1:** Welcome to the exciting world of Cybersecurity
- **Module 2:** The evolution of Cybersecurity
- **Module 3:** Protect against threats, risks, and vulnerabilities
- **Module 4:** Cybersecurity tools and programming language



Module 1

Welcome to the exciting world of Cybersecurity

Begin your journey into Cybersecurity!
You'll explore the Cybersecurity field,
and learn about the job responsibilities
of Cybersecurity professionals



Cybersecurity, or security, is the practice of ensuring confidentiality, integrity, and availability of information by

protecting networks, devices, people, and data from unauthorized access or criminal exploitation.

For example, requiring complex passwords to access sites and services improves confidentiality by making it much more difficult for a threat actor to compromise them.

Benefits of Cybersecurity-

- Security teams protect against external and internal threats.
- Security teams also ensure an organization meets regulatory compliance, or laws and guidelines that require the implementation of specific security standards.
- Security teams also maintain and improve business productivity by establishing a plan for business continuity,
- It also reduces expenses associated with risks, such as recovering from data loss or operational downtime, and potentially avoiding fines.
- Security maintains brand trust. If services or customer data are compromised, this can lower trust in the organization, damage the brand, and hurt the business in the long term.

Security professionals have two basic parts-

- **Operation side (For Analyst)**, which is responding to detections and doing investigations.
- **Project side (For Engineer)** where you're working with other teams to build new detections or improve the current detections.

Security professionals have two types of skills-

- **Technical skills:** Skills that require knowledge of specific tools, procedures, and policies.
- **Transferable skills:** Skills from other areas that can apply to different careers.

Technical skills-

- Programming languages
- Security information and event management (SIEM) tools
- Intrusion detection systems (IDSs)
- Threat landscape knowledge
- Incident response

Transferable skills-

- Communication
- Collaboration
- Problem-solving
- Time management
- Growth mindset
- Diverse perspectives.

Security analysts are responsible for monitoring and protecting information and systems.

- Protecting computers and network systems.
- Installing prevention software.
- Conducting periodic security audits.

Organizations are not the only ones that suffer during a data breach. Maintaining and securing user, customer and vendor data is an important part of preventing incidents that may expose people's **personally identifiable information (PII)** and **sensitive personally identifiable information (SPII)**

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If SPII is stolen, this has the potential to be significantly more damaging to an individual than if PII is stolen. PII and SPII data are key assets that a threat actor will look for if an organization experiences a breach. When a person's identifiable information is compromised, leaked or stolen, identity theft is the primary concern.

A **log** is a record of events that occur within an organization's systems. Logs help security professionals identify vulnerabilities and potential security breaches.

Module 2

The evolution of Cybersecurity

You will also understand how past and present cyber attacks have influenced the development of the security field and common attacks like malware and social engineering with their types and effects. In addition, you'll get an overview of the eight security domains.



Past Cybersecurity attacks

- **Brain virus** - In 1986, the Alvi brothers created the Brain Virus, although the intention of the virus was to track illegal copies of medical software and prevent pirated licenses, what the virus actually did was unexpected. Once a person used a pirated copy of the software, the virus-infected that computer. Then, any disk that was inserted into the computer was also infected. The virus spread to a new computer every time someone used one of the infected disks. Undetected, the virus spread globally within a couple of months.
- **Morris worm**-In 1988, Robert Morris developed a program to assess the size of the internet. The program crawled the web and installed itself onto other computers to all the number of computers that were connected to the internet and continued to re-install itself until the computers ran out of memory and crashed. About 6,000 computers were affected representing 10% of the internet at the time. This attack cost millions of **dollars** in damages due to business disruptions and the efforts required to remove the worm. After the Morris worm, Computer Emergency Response Teams, known as CERTs®, were established to respond to computer security incidents.

Attacks in the digital age

- **Love Letter attack-** In the year 2000, Onel De Guzman created the Love Letter malware to steal internet login credentials. Users received an email with the subject line, "I Love You." Each email contained an attachment labeled, "Love Letter for You." When the attachment was opened, the malware scanned a user's address book. Then, it automatically sent itself to each person on the list and installed a program to collect user information and passwords. Recipients would think they were receiving an email from a friend, but it was actually malware. The Love Letter ended up infecting 45 million computers globally and is believed to have caused over \$10 billion dollars in damages. The Love Letter attack is the first example of social engineering.
- **Equifax breach-** In 2017, attackers successfully infiltrated the credit reporting agency, Equifax. This resulted in one of the largest known data breaches of sensitive information. Over 143 million customer records were stolen, and the breach affected approximately 40% of all Americans. The records included personally identifiable information, Equifax settled with the U.S. government and paid over \$575 million dollars to resolve customer complaints and cover required fines

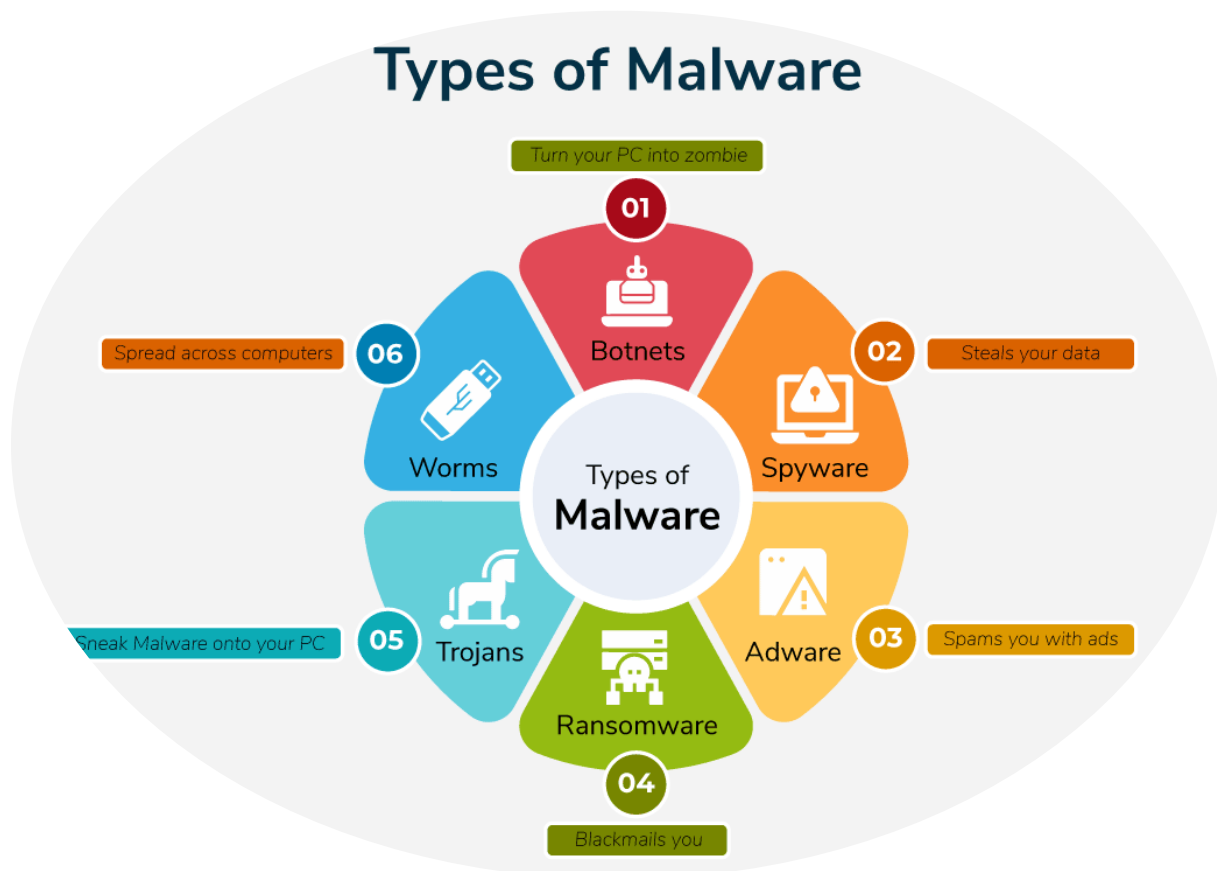
Common attacks and their effectiveness

Malware- It is software designed to harm devices or networks. Some of the most common types of malware attacks today include:

- 1. Botnet-** It is a self-replicating malware that spreads itself to other devices, creating a network of a *botnet*. Once infected, devices perform automated tasks commanded by the attacker. Botnets are often used in attacks.
- 2. Spyware:** Malware that's used to gather and sell information without consent. Spyware can be used to access devices. This allows threat actors to collect personal data, such as private emails, texts, voice and image recordings, and locations.
- 3. Adware-** It is software that displays or downloads unwanted advertisements, typically in the form of banners or pop-ups. It collects web browser history and cookies to target users with specific advertisements.
- 4. Ransomware:** A malicious attack where threat actors encrypt an organization's data and demand payment to restore access.
- 5. Trojan horse-** It is malicious software that appears legitimate to users. Once inside a device, the Trojan's payload -- malicious code -- is installed to facilitate the exploit. Trojans give attackers backdoor

access to a device, perform keylogging, install viruses or worms, and steal data.

- 6. Worms:** Malware that can duplicate and spread itself across systems on its own. In contrast to a virus, a worm does not need to be downloaded by a user. Instead, it self-replicates and spreads from an already infected computer to other devices on the same network.



Social Engineering - It is a manipulation technique that exploits human error to gain private information, access, or valuables. Social engineering attacks are related to the security and risk management domain. Some of the most common types of social engineering attacks today include:

- **Physical social engineering:** A threat actor impersonates an employee, customer, or vendor to obtain unauthorized access to a physical location.
- **Honeytrap attack-** It is a social engineering technique that specifically targets individuals looking for love on online dating websites or social media.
- **Scareware-** In this attack a scammer inserts malicious code into a webpage that causes pop-up windows. These windows will falsely alert you to a virus that's been installed on your system. You'll be told to purchase and download their security software and the scammers will either steal your credit card information, install real viruses on your systems.

- **Watering hole attack:** A threat actor attacks a website frequently visited by a specific group of user.
- **Phishing-** It is the use of digital communications to trick people into revealing sensitive data or deploying malicious software. Some of the most common types of phishing attacks today include:
 - **Spear phishing:** A malicious email attack that targets a specific user or group of users. The email seems to originate from a trusted source.
 - **Whaling:** A form of spear phishing. Threat actors target company executives to gain access to sensitive data.
 - **Vishing:** The exploitation of electronic voice communication to obtain sensitive information or to impersonate a known source.
 - **Smishing:** The use of text messages to trick users, in order to obtain sensitive information or to impersonate a known source.
 - **Business Email Compromise (BEC):** A threat actor sends an email message that seems to be from a known source to make a seemingly legitimate request for information, in order to obtain a financial advantage.

CISSP has defined eight security domains to organize the work of security professionals.



1. **Security and risk management** focuses on defining security goals and objectives, risk mitigation, compliance, business continuity and the law. For example, security analysts may need to update company policies related to private health information if a change is made to a federal compliance regulation
2. **Asset security** focuses on securing digital and physical assets. It's also related to the storage, maintenance, retention, and destruction of data. For example, old equipment is properly disposed of and destroyed, including any type of confidential information.

3. **Communication and network security** focuses on managing and securing physical networks and wireless communications. As a security analyst, you may be asked to analyze user behavior within your organization like users are connecting to unsecured wireless hotspots.
4. **Identity and access management** focuses on keeping data secure, by ensuring users follow established policies to control and manage physical assets, like office spaces, and logical assets, such as networks and applications. For example, as a security analyst, you may be tasked with setting up employees' keycard access to buildings.
5. **Security assessment and testing** focuses on conducting security control testing, collecting and analyzing data, and conducting security audits to monitor for risks, threats, and vulnerabilities. For example, access to payroll information is often limited to certain employees, so analysts may be asked to regularly audit permissions to ensure that no unauthorized person can view employee salaries.

6. **Security operations** focus on conducting investigations and implementing preventative measures. Imagine that you, as a security analyst, receive an alert that an unknown device has been connected to your internal network. You would need to follow the organization's policies and procedures to quickly stop the potential threat.
7. **Security architecture and engineering** focuses on optimizing data security by ensuring effective tools, systems, and processes are in place. As a security analyst, you may be tasked with configuring a firewall.
8. **Software development security** focuses on using secure coding practices, which are a set of recommended guidelines that are used to create secure applications and services. A security analyst may work with software development teams to ensure security practices are incorporated into the software development life-cycle.

Module 3

Protect against threats, risks, and vulnerabilities

You will learn about security controls and framework. You'll cover principles of the CIA triad, which are used to mitigate organizational risk and various National Institute of Standards and Technology (NIST) frameworks. In addition, you'll explore security ethics.

Security controls are safeguards designed to reduce specific security risks. For example, your company may have a guideline that requires all employees to complete a privacy training to reduce the risk of data breaches.

Security frameworks are guidelines used for building plans to help mitigate risks and threats to data and privacy. The purpose of security frameworks include protecting PII, securing financial information, identifying security weaknesses, managing organizational risks and aligning security with business goals.

Frameworks have four core components:

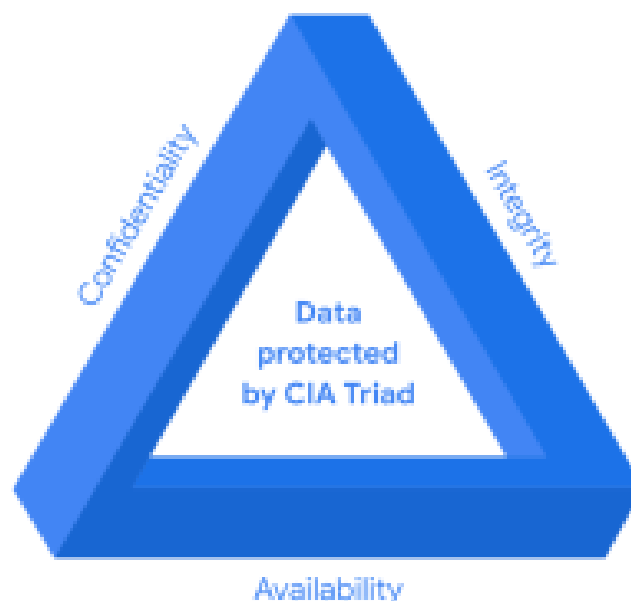
- Identifying and documenting security goals
- Setting guidelines to achieve security goals
- Implementing strong security processes
- Monitoring and communicating results

Examples of framework-

NIST Cybersecurity Framework (CSF) and the NIST Risk Management Framework (RMF). The National Institute of Standards and Technology (NIST) is a U.S.-based agency that develops multiple voluntary compliance frameworks that organizations

CIA triad (confidentiality, integrity, and availability)

is a model that helps inform how organizations consider risk when setting up systems and security policies. CIA is the three foundational principles used by cybersecurity professionals to establish appropriate controls that mitigate threats, risks, and vulnerabilities.



- **Confidentiality** means that only authorized users can access specific assets or data.
- **Integrity** means the data is correct, authentic, and reliable.
- **Availability** means data is accessible to those who are authorized to access it.

Ethics in cybersecurity

Security professionals must respect that privilege and act ethically at all times. Security ethics are guidelines for making appropriate decisions as a security professional.

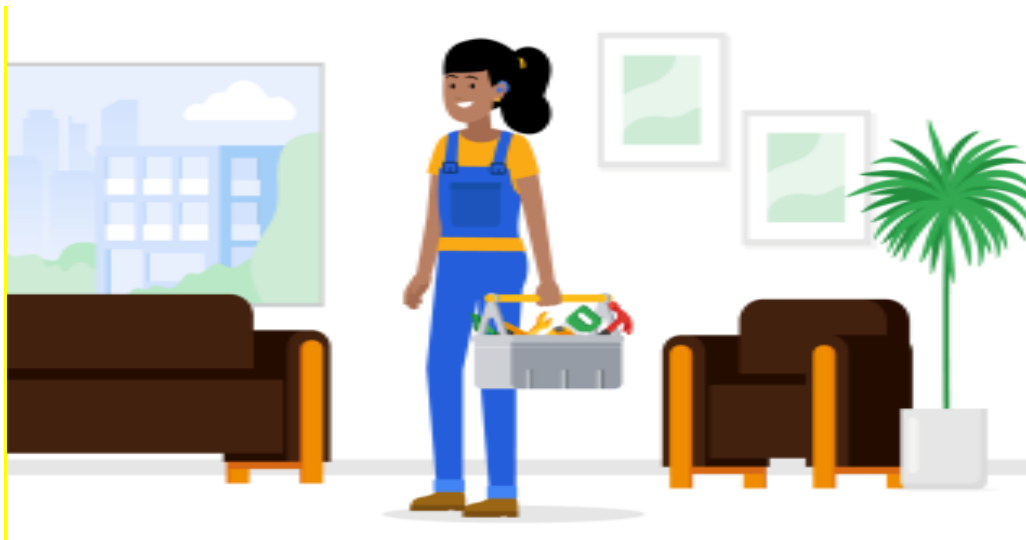
Ethical principles for mitigating risks-

- **Confidentiality** means that only authorized users can access specific assets or data. Confidentiality as it relates to professional ethics means that there needs to be a high level of respect for privacy to safeguard private assets and data.
- **Privacy protection** means safeguarding personal information from unauthorized use. As a security analyst, your role is to follow the policies and procedures of your company,
- **Laws** are rules that are recognized by a community and enforced by a governing entity.

Module 4

Cybersecurity tools and programming language

You'll discover common tools used by Cybersecurity analysts to identify and eliminate risk. You'll learn about security information and event management (SIEM) tools, playbook, network protocol analyzers, and programming languages such as Python and SQL.



Common Cybersecurity Tools

- **SIEM tool-** It collects and analyzes log data such as unusual login behavior, and helps to identify and analyze potential security threats, risks, and vulnerabilities more efficiently.

Commonly used SIEM tools:

- **Splunk** is a data analysis platform and Splunk Enterprise provides SIEM solutions. Splunk Enterprise is a self-hosted tool used to retain, analyze, and search an organization's log data.
 - **Google's Chronicle** is a cloud-native SIEM tool that stores security data for search and analysis. Cloud-native means that Chronicle allows for fast delivery of new features.
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- **Playbook-** It is a manual that provides details about any operational action, such as how to respond to an incident. Playbooks, which vary from one organization to the next, a guide analyst in how to handle a security incident before, during, and after it has occurred.
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- **Network protocol analyzer-** It is also known as a **packet sniffer**. It is a tool designed to capture and analyze data traffic in a network. This means that the tool keeps a record of all the data that a computer within an organization's network encounters. Common network protocol analyzers include **tcpdump** and **Wireshark**.

- **SQL** (Structured Query Language) - It is a programming language used to create, interact with, and request information from a database. An entry-level security analyst would use SQL to filter through the data points to retrieve specific information.
- **Python-** It is used to perform tasks that are repetitive and time-consuming and that require a high level of detail and accuracy. .
- **Operating system-** It is the interface between computer hardware and user. Linux, macOS, and Windows are operating systems.
- **Linux-** It is an open-source, or publicly available, operating system. Linux relies on a command line as the primary user interface. Linux itself is not a programming language, but it does allow for the use of text-based commands between the user and the operating system. A common use of Linux is examining logs to better understand what's occurring in a system.
- **Web vulnerability-** It is a unique flaw in a web application that a threat actor could exploit by using malicious code or behavior, to allow unauthorized access, data theft, and malware deployment.
- **Antivirus software-** It is a software program used to prevent, detect, and eliminate malware and viruses. It is also called **anti-malware**.

- **Intrusion detection systems (IDSs)**- It is used to monitor system activity and alerts for possible intrusions. They're a key tool that every organization uses to protect assets and data.
- **Encryption**- It makes data unreadable and difficult to decode for an unauthorized user; its main goal is to ensure confidentiality of private data. **Encryption** is the process of converting data from a readable format to a cryptographically encoded format.
- **Penetration testing**- It also called pen testing, is the act of participating in a simulated attack that helps identify vulnerabilities in systems, networks, websites, applications, and processes. It is a thorough risk assessment that can evaluate and identify external and internal threats as well as weaknesses.

Glossary terms from course

- **Adversarial artificial intelligence (AI):** A technique that manipulates artificial intelligence (AI) and machine learning (ML) technology to conduct attacks more efficiently.
- **Asset:** An item perceived as having value to an organization
- **Automation-** It is the use of technology to reduce human and manual effort in performing common and repetitive tasks. Automation also helps reduce the risk of human error.
- **CISSP:** Certified Information Systems Security Professional is a globally recognized and highly sought-after information security certification, awarded by the International Information Systems Security Certification Consortium.
- **Compliance:** It is the process of following rules and regulations to protect a company's digital information to avoid fines and security breaches.
- **External threat:** Someone outside of the organization trying to gain access to private information, networks or devices.
- **Firewall-** It is a device used to monitor and filter incoming and outgoing computer network traffic. Setting up a firewall correctly helps prevent attacks that could affect productivity.

- **Identity theft:** It is the act of stealing personal information to commit fraud while impersonating a victim and the primary objective of identity theft is financial gain.
- **Internal threat:** A current or former employee, external vendor, or trusted partner who poses a security risk.
- **Personally identifiable information (PII):** Any information used to infer an individual's identity PII includes someone's full name, date of birth, physical address, phone number, email address, internet protocol, or IP address and similar information.
- **Programming Language-** Programming can be used for searching data to identify potential threats or organizing and analyzing information to identify patterns related to security issues.
- **Security audit:** It is a review of an organization's security records,
- **Sensitive personally identifiable information (SPII):** A specific type of PII that falls under stricter handling guidelines and may include social security numbers, medical or financial information, and biometric data, such as facial recognition.
- **Threat:** Any circumstance or event that can negatively impact assets
- **Threat actor:** Any person or group who presents a security risk.